

Bayesian Networks I: Representation and Exact Inference

Lecture 19

Naive Bayes Was a Useful Bargain

Last time, multiple pieces of evidence became computationally simple when we assumed conditional independence:

$$P(O \mid e_1, \dots, e_n) \propto P(O) \prod_i P(e_i \mid O).$$

The bargain

Treat each feature as independent of the others once the class or hidden state is known.

The payoff

One prior plus one small likelihood model per feature keeps storage and inference cheap.

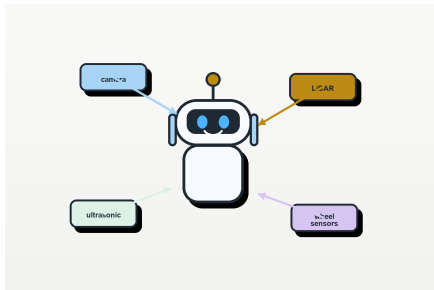
Key idea

Naive Bayes is fast because it makes a strong structural assumption about how evidence is related.

The Robot Now Has Multiple Sensors

Our delivery robot might use:

- ▶ a forward camera for visual object detection;
- ▶ LiDAR for range returns and point-cloud geometry;
- ▶ ultrasonic sensors for close-range distance;
- ▶ wheel and motion sensors to detect unexpected resistance.



Key idea

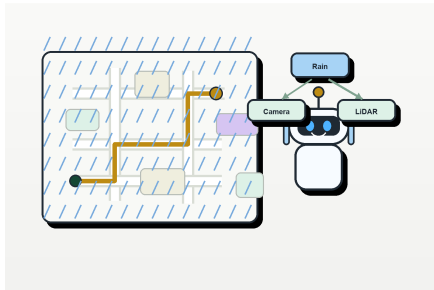
More sensors can improve perception—but only if we model how their evidence is related.

Multiple Sensors Share the Same World

Suppose we focus on two positive readings:

$$C = \text{camera}+, \quad L = \text{LiDAR}+.$$

- ▶ An actual obstacle can make both sensors fire.
- ▶ Rain can blur the camera and create noisy LiDAR returns.
- ▶ Dust, reflections, occlusion, or vibration can affect more than one sensor.
- ▶ The readings can be coupled even when there is no obstacle.



Naive Bayes Makes a Hidden Claim

Naive Bayes would use

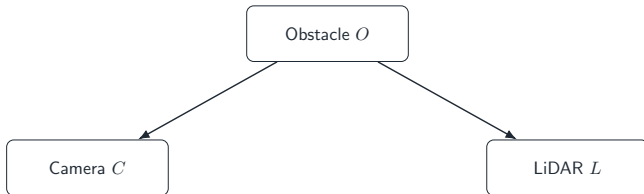
$$P(C, L \mid O) = P(C \mid O)P(L \mid O).$$

That equation asserts

$$C \perp\!\!\!\perp L \mid O.$$

In words

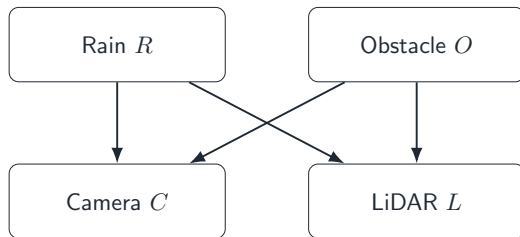
Once we know whether an obstacle exists, learning the camera result should tell us nothing more about the LiDAR result.



Key idea

If some unmodeled factor affects both sensors, this factorization double-counts shared evidence.

A Shared Cause Breaks the Naive Assumption



Rain is a common cause of sensor errors. If we omit it, C and L can remain correlated even after conditioning on $\neg O$.

Key idea

The fix is not “trust the sensors less.” The fix is to represent the dependency that creates the correlation.

Correlated Errors Can Make Us Overconfident

Using the model we will build today, when the path is clear:

$$P(C \mid \neg O) = 0.09, \quad P(L \mid \neg O) = 0.056.$$

Naive independence predicts

$$P(C, L \mid \neg O) \approx 0.09(0.056) = 0.00504.$$

But after accounting for rain,

$$P(C, L \mid \neg O) = 0.0108.$$

Key idea

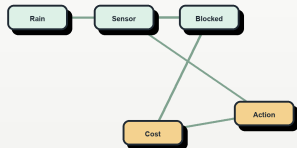
Treating correlated evidence as independent makes repeated agreement look more informative than it really is.

Bayesian Networks Make the Structure Explicit

Definition

A **Bayesian network** is a directed acyclic graph (DAG) whose nodes are random variables and whose edges define a factorization of their joint probability distribution.

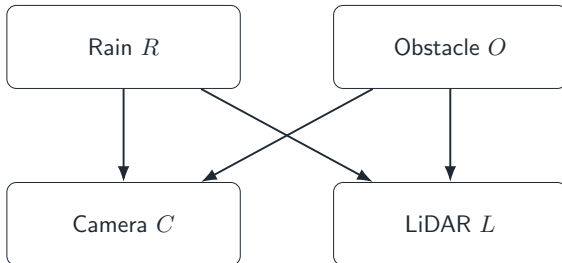
- ▶ **Graph:** which variables directly depend on which others.
- ▶ **CPTs:** numerical conditional probabilities at each node.
- ▶ **Factorization:** local terms multiply into the joint distribution.



Key idea

A Bayes net is both a probability model and a compact statement about conditional independence.

Our Robot Network



The graph says each sensor depends directly on obstacle state and rain—not on every variable in the model.

Key idea

A sparse graph states which dependencies are worth modeling and which dependencies we are willing to omit.

What Does an Arrow Mean?

Operational meaning

An edge $X \rightarrow Y$ means Y 's local probability table is conditioned on X .

For the robot,

$$O \rightarrow C \implies P(C \mid O, \dots),$$

and

$$R \rightarrow C \implies P(C \mid R, \dots).$$

Not automatically causation

A Bayesian network may have a causal interpretation, but the probabilistic definition only requires a valid DAG and factorization.

Key idea

Edges tell us what information the local conditional model needs. Missing edges are equally important.

The Graph Factorizes the Joint Distribution

For a Bayesian network,

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i)).$$

For our robot network,

$$P(O, R, C, L) = P(O)P(R)P(C \mid O, R)P(L \mid O, R).$$

- ▶ root nodes contribute priors;
- ▶ each non-root node depends only on its parents;
- ▶ multiplying the local factors recovers the probability of any complete assignment.

Key idea

Bayes nets replace one enormous joint table with a product of smaller local tables.

Structure Trades Parameters for Assumptions

Representation	Free probabilities	Assumption burden
Full joint: O, R, C, L	15	Almost none
Our Bayes net	10	$C \perp\!\!\!\perp L \mid O, R$
Naive sensor model	5	$C \perp\!\!\!\perp L \mid O$ and rain omitted

Four binary variables have 16 complete assignments, but a full joint needs only 15 free values because probabilities must sum to one.

Key idea

Model structure is compression: we save parameters and computation by declaring which dependencies we believe are unnecessary.

Root Nodes Need Priors

For today's toy model:

Obstacle O	Probability
true	0.10
false	0.90

Rain R	Probability
true	0.20
false	0.80

Because these nodes have no parents, their CPTs are simply prior distributions.

Key idea

A root node still has a CPT—it is just an unconditional probability table.

A CPT Encodes the Camera Model

O	R	$P(C = +)$
true	true	0.85
true	false	0.95
false	true	0.25
false	false	0.05

Interpret one row

$P(C = + \mid O = \text{true}, R = \text{true}) = 0.85$
means rain slightly reduces the camera's true-positive rate.

Interpret another

$P(C = + \mid O = \text{false}, R = \text{true}) = 0.25$
means rain also raises the false-positive rate.

Key idea

A CPT lets one sensor behave differently under different combinations of its direct parents.

The LiDAR Gets Its Own CPT

O	R	$P(L = +)$
true	true	0.90
true	false	0.98
false	true	0.20
false	false	0.02

- ▶ LiDAR is more reliable than the camera in this toy model.
- ▶ Rain still increases its false-positive rate.
- ▶ Camera and LiDAR can share the effect of rain without directly depending on each other.

Key idea

Separate local models can still coordinate globally because they share parent variables.

CPTs Have Their Own Scaling Problem

For a binary child with k binary parents,

$$\text{CPT parent rows} = 2^k.$$

Parents	Rows
2	4
5	32
10	1,024
20	1,048,576

Tradeoff

Adding an edge may improve model fidelity, but every extra parent can multiply the number of probabilities we must specify or learn.

Key idea

Bayes nets control the full-joint explosion only when each variable has relatively few direct parents.

Conditional Independence Is Now Local and Explicit

Our graph asserts

$$C \perp\!\!\!\perp L \mid O, R.$$

In words:

What the graph claims

If obstacle state and rain are known, then the camera reading provides no additional information about the LiDAR reading, and vice versa.

This is a much weaker—and more plausible—assumption than

$$C \perp\!\!\!\perp L \mid O.$$

Key idea

The network does not eliminate independence assumptions; it lets us state the right ones more precisely.

The Local Markov Property

Definition

Each node in a Bayesian network is conditionally independent of its non-descendants given its parents.

For the Camera node:

- ▶ $Parents(C) = \{O, R\}$;
- ▶ LiDAR is not a descendant of Camera;
- ▶ therefore $C \perp\!\!\!\perp L \mid O, R$.

Key idea

A graph is not just a picture; its topology encodes a family of conditional-independence statements.

Pattern 1: A Chain



$$A \rightarrow B \rightarrow C.$$

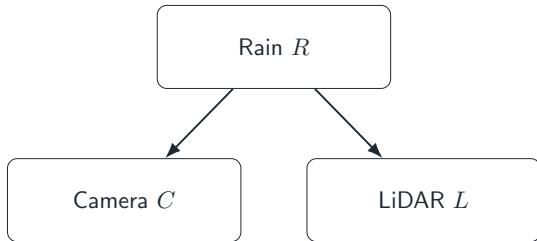
Condition on the middle

Once B is known, A provides no additional information about C .

Key idea

In a chain, observing the middle node blocks information flow between the ends.

Pattern 2: A Fork / Common Cause



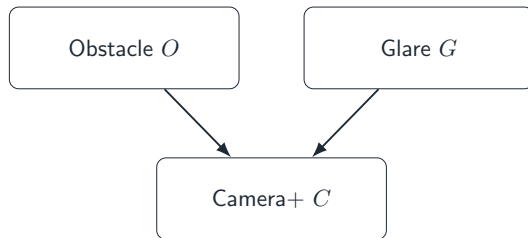
Before observing Rain

Camera and LiDAR can be correlated because they share a common cause.

After observing Rain

Conditioning on the common cause can block that dependence.

Pattern 3: A Collider



Counterintuitive rule

The two causes can be independent before observing *C*, but become dependent after we condition on their common effect.

Key idea

Colliders behave opposite to chains and forks: conditioning can open a dependency instead of blocking it.

Explaining Away

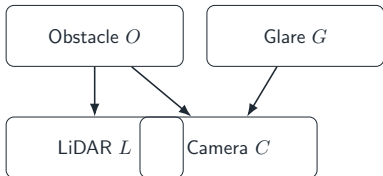
Suppose the camera fires. Two explanations are plausible:

- ▶ there really is an obstacle;
- ▶ sun glare caused a false positive.

Now LiDAR also fires.

Posterior effect

LiDAR evidence makes “obstacle” more plausible, so the camera reading needs “glare” less as an explanation.



Key idea

Conditioning can create dependence as well as remove it. Colliders are the key exception.

d-Separation: A Compact Independence Test

Path pattern	Condition on middle?	Effect
$A \rightarrow B \rightarrow C$	yes	blocks path
$A \leftarrow B \rightarrow C$	yes	blocks path
$A \rightarrow B \leftarrow C$	no	path blocked by default
$A \rightarrow B \leftarrow C$	yes, on B or descendant	opens path

Definition

Two variables are d-separated by the evidence when every path between them is blocked under these rules.

Key idea

The graph lets us answer many independence questions without multiplying a single probability.

Inference Means Asking the Network a Query

We want

$$P(O \mid C = +, L = +).$$

Query variable

O : Is there actually an obstacle?

Exact inference sums over every hidden state consistent with the evidence.

Evidence

$C = +, L = +$: both sensors fired.

Hidden variable

R : rain is not directly observed in this query.

Key idea

A Bayes-net query is probability calculation under the factorization encoded by the graph.

Exact Inference by Enumeration

Because Rain is hidden, sum over both rain states:

$$P(O \mid c, l) = \alpha \sum_r P(O)P(r)P(c \mid O, r)P(l \mid O, r).$$

The normalization constant α makes the two obstacle states sum to one.

Enumeration

Instantiate hidden variables, compute the joint probability for each assignment, sum the compatible assignments, then normalize.

Key idea

The BN factorization tells us exactly which local CPT entries to multiply for each hidden assignment.

Worked Query: Score the Obstacle Hypothesis

$$P(O, c, l) = P(O) \sum_r P(r) P(c \mid O, r) P(l \mid O, r).$$

Rain branch:

$$(0.20)(0.85)(0.90) = 0.153.$$

No-rain branch:

$$(0.80)(0.95)(0.98) = 0.7448.$$

Therefore

$$P(O, c, l) = 0.10(0.153 + 0.7448) = 0.08978.$$

Key idea

Hidden variables are not ignored; exact inference marginalizes over them.

Score the Clear Hypothesis, Then Normalize

$$P(\neg O, c, l) = 0.90 \left[(0.20)(0.25)(0.20) + (0.80)(0.05)(0.02) \right] = 0.00972.$$

Normalize:

$$P(O \mid c, l) = \frac{0.08978}{0.08978 + 0.00972} \approx 0.902.$$

Structured posterior

About 90.2%: rain can jointly explain two positive sensor readings.

Naive posterior

About 95.2% if the correlated evidence is incorrectly multiplied as independent.

Key idea

The difference is not a new Bayes rule; it comes from a better factorization of the evidence-generating process.

Variable Elimination Reuses the Factorization

Enumeration is correct, but larger networks repeatedly multiply the same subexpressions.

Eliminate Rain into a temporary factor:

$$f(O) = \sum_r P(r)P(c \mid O, r)P(l \mid O, r).$$

Then finish with

$$P(O \mid c, l) = \alpha P(O)f(O).$$

Definition

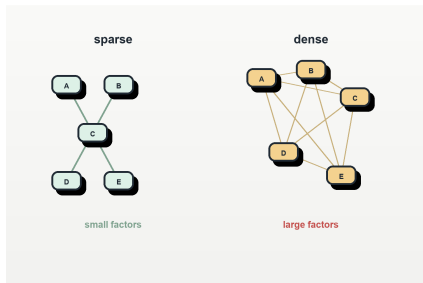
Variable elimination is an exact inference algorithm that multiplies relevant factors and sums out hidden variables in an organized order.

Key idea

Same probability calculation, less repeated work. On larger networks, elimination order can determine whether inference is practical.

Exact Does Not Mean Cheap

- ▶ Sparse graphs can make local CPTs small and inference efficient.
- ▶ Dense dependencies create large CPTs and large intermediate factors.
- ▶ Variable elimination can still be exponential in the network's induced width / treewidth.
- ▶ Choosing or learning the graph structure is itself a modeling problem.



Key idea

Bayesian networks trade unrestricted dependence for structured dependence. Good structure can buy tractability; dense structure can lose it.

Takeaways

- ▶ Naive Bayes is useful because of a strong conditional-independence assumption.
- ▶ Shared causes can correlate sensors and make naive evidence multiplication overconfident.
- ▶ A Bayesian network is a DAG plus local conditional probability tables.
- ▶ Its graph factorizes the joint distribution and encodes conditional independences.
- ▶ Chains, forks, and colliders behave differently under conditioning.
- ▶ Exact inference marginalizes hidden variables; variable elimination organizes that computation.
- ▶ The central tradeoff is model fidelity versus parameter count and computational cost.

Key idea

Today: represent structured uncertainty exactly. Next: what happens when exact inference itself becomes too expensive?